MSAJCE — Mohamed Sathak A.J. College of Engineering

Technology Centre — Programme & Course Details

1. Technology Centre Overview and Purpose

Technology Centre — MSAJCE

In pursuit of MSAJCE's vision and mission to make students deployable and equip them with upcoming emerging industrial demands, various Technology Centres were established in the year 2019-2020 to educate, train and up-skill students in niche technologies. The courses and training are designed based on the recommendations of AICTE and India Skill Report 2020. The Technology Centers, established with state-of-art facilities, provide an ambient learning environment for research and innovation. Each centre is collaborated with industry/government organizations to provide certifications, onsite/offsite projects and internships.

Industry Partners

Automation Anywhere, Godrej Inc, Ford, Palo Alto Systems, Amazon Web Services, National Instruments, CISCO Academy, Altair, Open Wave Computing, Levergent Technologies, Propeller Technologies.

MECH & CIVIL Technology Centres

S.No	Name of Centre
1	Centre for 3D Printing and Additive Manufacturing
2	Centre for CAE
3	Centre for Industrial Robotics
4	Centre for NDT – Non Destructive Testing
5	Centre for Building Information Modelling

ECE & EEE Technology Centres

S.No	Name of Centre
1	Centre for Embedded & IoT
2	Centre for UAV/Drone Technology
3	Godrej Disha Skill Development Centre
4	Centre for Bosch Industry Institute Collaboration
5	Centre for E-Mobility
6	Centre for Renewable Energy
7	APJ Abdul Kalam Innovation Centre

CSE & IT Technology Centres

S.No	Name of Centre
1	Centre for Bot Lab & Robotics Process Automation
2	Centre for Cisco Networking Academy
3	Centre for Code Tantra – Coding Learning Centre
4	Centre for Gaming and AR/VR

5	Centre for Artificial Intelligence and Machine Learning
6	Centre for Block Chain Technology
7	Centre for Mobile and Web Application Development

2. RPA Bot Lab — Robotics Process Automation (Automation Anywhere)

Robotics Process Automation (RPA) — Partnered with Automation Anywhere

About the Centre

MSAJCE has partnered with Automation Anywhere, the market leaders in Robotic Process Automation (RPA), to establish the AAU powered Bot Lab. The AAU powered Bot Lab provides training in RPA, which is simple and powerful automation software used to create software robots that can automate any business process.

Objectives — Students will be able to:

- Explain in detail about the Automation Anywhere Enterprise Platform, Architecture and its Components.
- Provide in-depth knowledge about the various features and functionalities of AA software, Runtime AA Client, and Web Control Room.
- Create Bots using different types of Recorders, Editors, and Basic Commands using Automation Anywhere platform.
- Apply bot creation methodologies for developing bots to solve any real world problems.

Outcomes — Students will be able to:

- Automate any sort of manual or repetitive human tasks by using the Bot Development tool.
- Develop and deploy software bots which can perform automation tasks 24/7 without interruption.
- Create and deploy bots which significantly improves the process and productivity of any industrial, educational, government and private organizations.
- Build IQ bots with artificial intelligence capable of solving complex real world problems.

Benefits

- Access to the Automation Anywhere Enterprise Platform and hands-on training.
- Globally-recognized certification, internships and placement assistance across industries.
- Opportunity to build your own bots and upload it on Automation Anywhere Bot Store, the world's leading marketplace for bots and Digital Workers.

Course Offered

Robotic Process Automation

Certifications

- Automation Anywhere Certified Essential RPA Professional
- Automation Anywhere Certified Advanced RPA Professional
- Automation Anywhere Certified Master RPA Professional

Syllabus

1. Introduction to RPA & Bot Creation (6 Hours)
2. Bot Creator (9 Hours): Introduction – Recorders – Smart Recorders – Web Recorders – Screen Recorders – Task Editor – Variables – Command Library – Loop Command – Excel Command – Database Command – String Operation Command – XML Command – Terminal Emulator Command – PDF Integration Command – FTP Command – PGP Command – Object Cloning Command – Error Handling Command – Manage Windows Control Command – Workflow Designer – Report Designer – Best Practices – Summary

3. Meta Bot and Bot Insight (6 Hours): Introduction – MetaBot Designer – MetaBot With AI Sense – Bot Insight – Transactional Analytics – Operational Analytics

4. Assessment and Essential Certification (6 Hours)

5. List of Experiments (12 Hours): Software Installation (AA Control Room, SQL Server, AA Client) – Bot Creation using Smart/Web/Screen Recorders – Loop Command bots – Database automation bots – Excel automation bots – PDF Integration bots – Error handling bots – Object Cloning – FTP and PGP execution – MetaBot with AI Sense.

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3. CISCO Networking Academy Centre — Objectives, Outcomes and Certifications

CISCO Network Academy Centre

About the Centre

MSAJCE has partnered with CISCO, the world leaders in networking, to establish the Networking Academy (NetAcad). Cisco Networking Academy is an IT skills and career building program providing unique hands-on learning in networking, security, and cloud technologies using simulation tools. Professional certification is recognized by employers worldwide. The Academy provides career pathways to an estimated 8 million networking jobs worldwide through Netacad.

Objectives — Students will be able to:

- Understand the architecture, structure, functions, components, and models of the Internet and other computer networks using OSI and TCP layered models.
- Learn the principles and structure of IP addressing and fundamentals of Ethernet concepts, media, and operations.
- Use Packet Tracer (PT) to analyze protocol and network operation and build small networks in a simulated environment.

Outcomes — Students will be able to:

- Build simple LANs, perform basic configurations for routers, switches, and implement IP addressing schemes.
- Configure and troubleshoot routers, switches and resolve VLAN routing issues in IPv4 and IPv6 networks.
- Configure WAN technologies and network services required by converged applications in a complex network.
- Acquire knowledge to use simulation tools to design an enterprise network and evaluate design alternatives.
- Address IP addresses assignment, internetworking equipment selection and network performance issues.

Courses Offered

Network Essentials | Mobility Fundamentals | CCNA: Introduction to Networks | CCNA: Switching, Routing, and Wireless Essentials | CCNA: Enterprise Networking, Security and Automation | Networking Courses | Security Courses | IoT & Data Analytics Courses | OS and IT Courses | Programming Courses | Entrepreneurship

Certifications

- CISCO Certified Network Associate (CCNA)
- Implementing CISCO Enterprise Network Core Technologies

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4. CISCO CCNAv7 — Introduction to Networks (Full Syllabus)

Total Hours: 70 | Source: CISCO Networking Academy

24 hands-on and paper-based labs using Packet Tracer Tool, Cisco 4221 routers and 2960 switches.

No.	Topic	Hours
1	Networking Today	3
2	Basic Switch and End Device Configuration: Cisco IOS Access – IOS Navigation – Command Structure – Basic Device Con	5
3	Protocol and Models: The Rules – Protocol Suites – Standards Organizations – Reference Models – Data Encapsulation – D	3
4	Physical Layer: Purpose – Physical Layer Characteristics – Copper Cabling – UTP Cabling – Fiber-Optic Cabling – Wireless	3
5	Number Systems: Binary Number System – Hexadecimal Number System	1
6	Data Link Layer: Purpose – Topologies – Data Link Frame	3
7	Ethernet Switching: Ethernet Frame – Ethernet MAC Address – The MAC Address Table – Switch Speeds and Forwarding M	4
8	Network Layer: Network Layer Characteristics – IPv4 and IPv6 Packet – Router Routing Table	5
9	Address Resolution: MAC and IP – ARP – Neighbor Discovery	3
10	Basic Router Configuration: Configure Initial Router Settings – Configure Interfaces – Configure the Default Gateway	6
11	IPv4 Addressing: IPv4 Address Structure – Unicast/Broadcast/Multicast – Types of IPv4 Addresses – Network Segmentation	6
12	IPv6 Addressing: IPv6 Addressing – Address Types – GUA and LLA Static Configuration – Dynamic Addressing – IPv6 Mult	4
13	ICMP: ICMP Messages – Ping and Traceroute Testing	3
14	Transport Layer: Transportation of Data – TCP Overview – UDP Overview – Port Numbers – TCP Communication Process –	5
15	Application Layer: Application, Presentation, and Session – Peer-to-Peer – Web and Email Protocols – IP Addressing Servic	5
16	Network Security Fundamentals: Security Threats and Vulnerabilities – Network Attacks – Attack Mitigation – Device Securit	5
17	Build a Small Network: Devices – Small Network Applications and Protocols – Scale to Larger Networks – Verify Connectivit	6

5. Centre for 3D Printing and Additive Manufacturing

Centre for 3D Printing and Additive Manufacturing

Objectives — Students will be able to:

- Gain knowledge and skills related to 3D printing technologies.
- Learn the selection of material, equipment and development of a product for Industry 4.0 environment.
- Understand various software tools, processes and techniques for digital manufacturing.
- Apply these techniques into various applications.

Outcomes — Students will be able to:

- Develop CAD models for 3D printing.
- Import and export CAD data and generate .stl files.
- Select a specific material for a given application.
- Select a 3D printing process for an application.
- Produce a product using 3D Printing or Additive Manufacturing (AM).

Syllabus Modules

1. 3D Printing (Additive Manufacturing) — 6 Hours: Introduction, Process, Classifications, Advantages, Additive vs Conventional Manufacturing, Applications.
2. CAD for Additive Manufacturing — 4 Hours: CAD Data formats, Data translation, Data loss, STL format.
3. Additive Manufacturing Techniques — 10 Hours: Stereo-Lithography, LOM, FDM, SLS, SLM, Binder Jet technology. Process parameters, Process Selection. Application Domains: Aerospace, Electronics, Healthcare, Defence, Automotive, Construction, Food Processing, Machine Tools.
4. Materials — 4 Hours: Polymers, Metals, Non-Metals, Ceramics. Various forms of raw material (Liquid, Solid, Wire, Powder). Powder Preparation and their desired properties. Support Materials.

5. Additive Manufacturing Equipment — 4 Hours: Process Equipment Design and parameters, Governing Bonding Mechanism, Common faults and troubleshooting, Process Design.

6. Post Processing — 1 Hour: Support Removal, Sanding, Acetone treatment, Polishing.

7. Product Quality — 1 Hour: Inspection and testing, Defects and their causes.

List of Practicals

- 3D Modelling of a single component
- Assembly of CAD modelled Components
- Exercise on CAD Data Exchange
- Generation of .stl files
- Identification of a product for AM and its process plan
- Printing on an available AM machine
- Post processing of additively manufactured product
- Inspection and defect analysis
- Comparison of AM product with conventional manufactured counterpart

Recommended Textbooks

- Lan Gibson, David W. Rosen and Brent Stucker, 'Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing', Springer, 2010.
- Andreas Gebhardt, 'Understanding Additive Manufacturing', Hanser Publisher, 2011.
- Khanna Editorial, '3D Printing and Design', Khanna Publishing House, Delhi.
- CK Chua, Kah Fai Leong, '3D Printing and Rapid Prototyping', World Scientific, 2017.

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6. Amazon Web Services (AWS) & Cloud Computing Centre

Amazon Web Services (AWS) & Cloud Computing Centre

Industry Partner: Amazon Web Services (AWS)

This centre is part of MSAJCE's broader Technology Centre ecosystem established in 2019-2020 under AICTE and India Skill Report 2020 recommendations.

Objectives — Students will be able to:

- Understand the fundamentals of building IT infrastructure on the AWS platform.
- Acquire knowledge of various cloud services provided by Amazon Web Services (AWS).
- Learn how to create and configure AWS services and integrate various AWS services together to present a holistic cloud solution.
- Learn how these services work together to build fault tolerant and highly available applications.
- Understand best practices and design patterns to architect optimal IT solutions on AWS.

Outcomes — Students will be able to:

- Create and configure AWS services and integrate various AWS services together to present a holistic cloud solution.
- Build fault tolerant and highly available applications on cloud.
- Apply design patterns to architect optimal IT solutions on AWS.

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7. Cyber Security Academy — Palo Alto Systems

Cyber Security Academy — Palo Alto Systems

Objectives — Students will be able to:

- Essentials I & II: Learn general security concepts for maintaining a secure network, the nature and scope of today's cybersecurity challenges, strategies for network defense and next-generation cybersecurity solutions.
- Cyber Security Foundation: Learn fundamentals of cybersecurity, identify concepts to recognize and potentially mitigate attacks against enterprise networks and mission critical infrastructure.
- Gateway I & II: Learn fundamental tenets of networking and concepts for maintaining a secure network computing environment.

Outcomes — Students will be able to:

- Essentials I: Formulate industry-standard designs to protect infrastructure against cybersecurity threats; apply user, application, and content ID filtering; deploy security methodologies for a secure network environment.
- Cyber Security Foundation: Apply fundamentals to identify and mitigate attacks against enterprise networks.
- Gateway I: Demonstrate knowledge of interconnected technology; identify systems needing protection; apply physical, logical, and virtual addressing through subnet mask schemes.
- Gateway II: Examine cybersecurity landscapes, attack threat vectors, exposure, vulnerabilities, and risk factors; apply knowledge to plan, design, implement, troubleshoot, and maintain network infrastructure.
- Essentials II: Assess and harden endpoints; use advanced malware research and analysis; examine mobile and cloud-based connection technologies.

Detailed Syllabus

1. Fundamentals of Cyber Security (4 Hours): Security trends – Legal, Ethical and Professional Aspects – Need for Security at Multiple levels – Security Policies – Model of network security – OSI security architecture.
2. Cyber Security Concepts (4 Hours): Classical encryption (substitution, transposition, steganography) – Modern cryptography – CIA, Risks, Breaches, Threats, Attacks, Exploits – Information Gathering (Social Engineering, Foot Printing & Scanning).
3. Lab Session 1 (3 Hours): Tools like nmap, zenmap, Port Scanners, Network Scanners.
4. Security Algorithms (4 Hours): Asymmetric key Cryptography – Message Authentication – Digital Signatures – Applications of Cryptography – Firewalls – Types of Firewalls – User Management – VPN Security.
5. Lab Session 2 (3 Hours): Calculate message digest using SHA-1 algorithm.
6. Infrastructure Based Security (6 Hours): Network packet sniffing – Network Design Simulation – DOS/DDOS attacks – Asset Management and Audits – Vulnerabilities and Attacks – Intrusion Detection and Prevention – HIPS – Security Information Management – Network Session Analysis – System Integrity Validation.
7. Lab Session 3 (3 Hours): Implement sniffing using tools.
8. Lab Session 4 (3 Hours): Demonstrate IDS using Snort.
9. Lab Session 5 (3 Hours): Automated Attack and Penetration Tools using N-Stalker.
10. Malware (6 Hours): Explanation of Malware – Types: Virus, Worms, Trojans, Rootkits, Robots, Adware, Spywares, Ransomware, Zombies – OS Hardening – Malware Analysis.
11. Lab Session 6 (6 Hours): Building Trojans – Defeating Malware – Rootkit Hunter.

Sources: Anna University CNS Lab syllabus | AICTE emerging technology syllabus | IGNOU security certificate.

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8. Embedded & IoT Centre

Embedded & IoT Centre

Objectives

- Learn Smart Objects and IoT Architectures.
- Learn various IoT-related protocols.
- Build simple IoT Systems using Arduino and Raspberry Pi.
- Learn data analytics and cloud in the context of IoT.
- Develop IoT infrastructure for popular applications.

Outcomes

- Understand the concept of IoT.
- Analyze various protocols for IoT.
- Design a PoC of an IoT system using Raspberry Pi/Arduino.
- Apply data analytics and use cloud offerings related to IoT.
- Analyze applications of IoT in real time scenarios.

Syllabus

1. Introduction to IoT (8 Hours): Architectural Overview – Design principles – IoT Applications – Sensing – Actuation – Basics of Networking – M2M and IoT Technology Fundamentals – Devices and gateways – Data management – Business processes in IoT – Everything as a Service (XaaS) – Role of Cloud in IoT – Security aspects in IoT.

2. Elements of IoT (9 Hours): Hardware Components (Arduino, Raspberry Pi, Communication, Sensing, Actuation, I/O Interfaces) – Software Components – Programming APIs using Python/Node.js/Arduino – Communication Protocols: MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.

3. IoT Application Development (18 Hours): Solution framework for IoT applications – Implementation of Device integration – Data acquisition and integration – Device data storage (unstructured data on cloud/local server) – Authentication and authorization of devices.

4. IoT Case Studies (10 Hours): Mini projects in Industrial automation, Transportation, Agriculture, Healthcare, Home Automation.

List of Practicals (15 Experiments)

- Familiarization with Arduino/Raspberry Pi and software installation.
- Interface LED/Buzzer with Arduino/Raspberry Pi — turn ON LED for 1 sec every 2 seconds.
- Interface Push button/Digital sensor (IR/LDR) — turn ON LED on detection.
- Interface DHT11 sensor — print temperature and humidity readings.
- Interface motor using relay — turn ON motor on button press.
- Interface OLED — print temperature and humidity.
- Interface Bluetooth — send sensor data to smartphone.
- Interface Bluetooth — turn LED ON/OFF via smartphone.
- Upload temperature and humidity data to ThingSpeak cloud.
- Retrieve temperature and humidity data from ThingSpeak cloud.
- Install MySQL database on Raspberry Pi and perform basic SQL queries.
- Publish temperature data to MQTT broker.
- Subscribe to MQTT broker for temperature data.
- Create TCP server on Arduino/Raspberry Pi — respond with humidity data.
- Create UDP server — respond with humidity data to UDP client.

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9. e-Yantra IIT Bombay Remote Centre & Code Tantra Coding Learning Centre

e-Yantra — IIT Bombay Remote Centre

e-Yantra is an IIT Bombay initiative. MSAJCE is a Remote Centre for e-Yantra under IIT Bombay.

Objectives — Students will be able to:

- Learn the function of basic components of robots.
- Study various types of end effectors and sensors.
- Impart knowledge in Robot kinematics and Programming.
- Learn Robot safety issues and economics.

Outcomes — Students will be able to:

- Understand embedded systems and robotics technology.
- Design, develop, program and test robots for various applications.
- Participate in national and international robotics competitions confidently.
- Use robots to solve real life problems.

Code Tantra — Coding Learning Centre

Industry Partner: Code Tantra

Objectives — Students will be able to:

- Understand the syntax and semantics of various programming languages (C, C++, Java, Python).
- Get a new perspective of problem solving by choosing appropriate programming domains for software product development.
- Make students competent data analyzers or data scientists by learning Big Data and Hadoop.
- Increase the ability of program testing and debugging using Selenium testing tool.
- Prepare students for competitive based learning by participating in global level coding contests.

Outcomes — Students will be able to:

- Design and implement software applications using C, C++ basic constructs.
- Develop software products for industries in various programming platforms.
- Understand the key issues in big data management and its associated applications in intelligent business and scientific computing.
- Test and debug various software products effectively.
- Develop real-time web and mobile applications.

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10. Gaming & AR/VR Centre

Gaming & AR/VR Centre

Objectives — Students will be able to:

- Learn technical and design aspects of AR and VR.
- Create 3D design, Visualization and Animation for various objects and scenes.
- Apply virtual and augmented reality methods, tools and equipment in industry, education and social life.
- Provide an overview of the concepts and working of AR-VR techniques.

Outcomes — Students will be able to:

- Build basic applications considering technical and design aspects of AR and VR.
- Develop PC/Mobile-based VR applications.
- Develop 3D Visualization and Visualization projects.
- Design applications in Gaming, Architecture, Manufacturing, Retail, Healthcare, etc.

Syllabus Modules

1. Introduction to Virtual Reality (6 Hours): Fundamental Concept and Components of VR – Primary Features and Present Development – Computer graphics – Real time computer graphics – Flight Simulation – Benefits of VR – Historical development – 3D Computer Graphics: Virtual world space, positioning the virtual observer, perspective projection, human vision, stereo perspective projection, 3D clipping, Colour theory, Simple 3D modelling, Illumination models, Reflection models, Shading algorithms, Radiosity, Hidden Surface Removal, Realism-Stereographic image.

2. Interactive Techniques in VR (6 Hours): From 2D to 3D – 3D space curves – 3D boundary representation – Geometric Transformations – Frames of reference – Modeling transformations – Instances – Picking – Flying – Scaling the VE – Collision detection – Generic VR system: Virtual environment, Computer environment, VR technology, Model of interaction, VR Systems.

3. Visual Computation in VR (6 Hours): Animating the Virtual Environment – Dynamics of numbers – Linear and Nonlinear interpolation – Animation of objects – Shape and object inbetweening – Free-form deformation – Particle system – Physical Simulation: Objects in gravitational field, Rotating wheels, Elastic collisions, Projectiles, Simple pendulum, Springs, Flight dynamics.

4. Augmented and Mixed Reality (6 Hours): Taxonomy, technology and features of AR – Difference between AR and VR – Challenges with AR – AR systems and functionality – AR methods – Visualization techniques – Wireless displays in educational AR – Mobile projection interfaces – Marker-less tracking – Enhancing interactivity in AR – Evaluating AR systems.

5. Multiple Models of Input and Output Interface in VR (6 Hours): Human factors – VR Hardware: Head-coupled displays, Acoustic hardware, Integrated VR systems – VR Software: Modeling virtual world, Physical simulation, VR toolkits, Introduction to VRML – Input: Tracker, Sensor, Digital Glove, Movement Capture, Video-based Input, 3D Menus, 3D Scanner – Output: Visual/Auditory/Haptic Devices.

6. Application of VR in Digital Entertainment (6 Hours): VR in Film & TV Production – VR in Physical Exercises and Games – Demonstration of Digital Entertainment by VR.

Practicals (9 Hours)

- Architecture design using VR
- CRO experiments using VR
- Qualitative analysis in Chemistry using VR
- Engine assembly/disassembly using VR
- Human anatomy exploration
- Simulation of blood circulation
- Simulation of Flight/Vehicle/Space Station
- Building Electronic circuits using VR
- Developing Virtual Classroom with multiplayer

Resources from: AICTE | IIT Varanasi, India

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11. Bentley's BIM Centre and Centre for CAE — Altair & CATIA

Bentley's Building Information Modelling (BIM) Centre

Objectives

- Gain basic theoretical knowledge and practical skills of BIM techniques.
- Interpret 2D & 3D drawings and structural analysis.
- Understand regulations as per National Building Code.
- Prepare students for actual work situations through practical training.

Outcomes

- Prepare a component to meet desired needs within realistic constraints (economic, environmental, social, ethical, sustainability).
- Create 2D drawings of building, industrial structures and various framed structures using AUTOCAD software.
- Create and simulate a virtual building model using REVIT ARCHITECTURE software with intelligent building elements.
- Design and analyse different types of structures (liquid retaining structures, bridge components, retaining wall, industrial and commercial structures) in 3D form using STAAD.Pro software to predict performances before construction.

Software Used: AUTOCAD, REVIT ARCHITECTURE, STAAD.Pro

Centre for CAE — Supported by Altair & CATIA

Industry Partner: Altair

Objectives

- Understand and interpret drawings of machine components.
- Prepare assembly drawings using standard CAD packages.
- Gain practical experience in handling 2D drafting and 3D modeling software systems.
- Comprehend, design, analyze, and suggest novel materials, products and solutions for contemporary manufacturing issues.

Outcomes

- Develop CAD models for Hypermesh.
- Import and export CAD data and generate .hm files.
- Attain accurate and high quality meshes in 1D, 2D and 3D elements.
- Carry out Linear Static Analysis.
- Carry out Dynamic Analysis.

Software Used: Altair HyperMesh, CATIA

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12. Communication Language Learning Centre

Communication Language Learning Centre

Objective

- Acquire knowledge in foreign languages and become proficient in communication skills — Listening, Speaking, Reading and Writing (LSRW) — to achieve employability standard at international level.

Outcomes — Students will be able to:

- Become expert bilinguals to learn work ethic and business etiquette in career opportunities.
- Gain deeper knowledge and higher understanding of foreign languages.
- Boost professional value in Business, Social Science, Humanities, Technology and Science.
- Get exposure to culture ingrained in the people of various countries.
- Improve higher thinking skills, new learning strategies, and exposure to advanced technologies.
- Grow as leaders in many leading sectors to drive for innovation.

Focus: Language Skills (LSRW — Listening, Speaking, Reading, Writing), Foreign Language Proficiency, International Employability, Business Communication.

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13. RAISE Centre of Excellence — MIT Square London

RAISE Centre of Excellence (CoE) — Powered by MIT Square, London

Established: June 2022

About MIT Square Services Private Limited

MIT Square is an ISO Certified Company headquartered in Bangalore (Silicon Valley of India) and registered office in London (Silicon Valley of UK). It is a registered MSME recognised by the Ministry of MSME, Govt. of India and recognised by the Ministry of Corporate Affairs, Govt. of UK. Network offices in USA, Australia and Singapore. MIT stands for 'Management and Innovation for Transformation'. Square represents 'People + Process + Product + Profit' — tagline: 'We transform your life'. Core technical team from IITs and Universities in London.

MIT Square is a globally trusted brand in EduTech, HealthTech, AgroTech and GreenTech supporting: incubation/start-up, business operations, talent enablement (training & learning), talent acquisition, innovations, patent filing, product design & development, manufacturing and supply of technological and non-technological products.

About RAISE

RAISE = R – Research | A – Arts & Applications | I – Innovation & Intelligence | S – Science & Technology | E – Entrepreneurship

The CoE for IoT at MSAJCE campus was established in June 2022 through MoU with MIT Square London. Vision: Enable the institution as an innovation hub in emerging technologies.

Key Initiatives — i-10 Model

- Innovation & Research
- Ideas to Prototypes & Products
- Inductive Programs
- Internships & Talent Transformation
- International Publications
- I-Centric Events
- IP Rights & Protection
- Intrapreneurship & Entrepreneurship
- Institution & Industry Connects
- Incubation, Investments & Fundings

Specific Initiatives

Abroad Studies | Abroad Traineeships | Abroad Internships | Abroad Placementships | Abroad Mentorships | Abroad Fellowships | Abroad Inventorships | Startup Founderships | Disruptive Technologies | Product Development | CSR Activities

Detailed Objectives

- Create innovative applications and domain capability for Smart City, Smart Health, Smart Manufacturing, Smart Agriculture.
- Build industry capable talent, startup community and entrepreneurial ecosystem.
- Provide ecosystem for innovation and entrepreneurship.
- Energize research mind-set and reduce costs in R&D; with neutral, interoperable, multi-technology stack laboratory facilities.
- Reduce import dependency on technology components and promote indigenization.
- Position the institution as a provider of end-to-end solutions in education and technology.
- Provide training and learning (including internships) both online and offline.
- Provide environment for product creation, testing, validation and incubation.
- Provide learning in breakthrough technologies: AR/VR, Data Analytics, AI.

14. RAISE Centre — Benefits, Nodal Centre Details and Contact Information

RAISE Centre — Benefits for Students and Nodal Centre Details

Benefits of RAISE CoE

- A great learning and leadership opportunity.
- Become a coding guru.
- Spread Techno-Entrepreneurship Culture within the Campus through events, workshops and hackathons.
- Internship Opportunities.
- Opportunity to work in startups or prestigious companies.
- Opportunity to study abroad.
- Opportunity to intern or work full-time abroad.
- Personalised LinkedIn recommendation.
- Letter of Recommendation after successful Internship/techno programs completion.

MSAJCE as RAISE Nodal Centre — Benefits

- A fully Techno-Entrepreneurship Innovation lab with hardware and software tools with licenses for development and research projects.
- Recognition as one of the top National Academies for the innovation lab.
- Essential curriculum and resource kits for comprehensive training and industry readiness.
- Scalable infrastructure to extend training to corporates and professionals from other industries.
- Up to 10 years assured training support from MIT Square to empower faculty, research scholars and students.
- Based on NEP 2020, support provided in regional languages as well.

Government Initiatives Supported

Digital India | Make in India | Startup India | Standup India | Skill India | Self-reliant India (Atmanirbhar Bharat)

MSAJCE Contact Information

Institution	Mohamed Sathak A.J. College of Engineering (MSAJCE)
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Email	msajce.office@gmail.com
Grievance Email	grievance@msajce-edu.in
Website	www.msajce-edu.in
Affiliation	Affiliated to Anna University Chennai
Approval	Approved by AICTE New Delhi
Trust	Under the aegis of Mohamed Sathak Trust | Government of Tamil Nadu approved